

DAY 07: GIT WORKFLOW, DEPLOYMENT CHALLENGES, DOCKERFILE & DOCKER BASICS

Cloud Computing & DevOps Master Curriculum

1. Challenges with Traditional Deployment

Traditional software deployment often suffers from the *'It works on my machine'* problem due to differences between development, testing, and production environments.

- OS and dependency version mismatches across servers.
- High resource overhead from heavy hypervisor Virtual Machines.
- Slow deployment cycles and tedious manual server provisioning.

2. Containerization vs Virtualization

Containerization packages an application and all its dependencies together into an isolated user-space container that shares the host operating system kernel.

Unlike Virtual Machines, containers do not require a full guest OS, making them lightweight (MBs instead of GBs), blazing fast to start (seconds), and portable across any cloud or machine.

3. Dockerfile Directives & Key Commands

A **Dockerfile** is a text file containing sequential instructions to build a Docker Image.

```
FROM node:18-alpine
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
CMD ["node", "server.js"]
```

Essential Docker CLI Commands:

```
# Build Docker image
docker build -t myusername/my-web-app:v1.0 .

# Run container on port 8080
docker run -d -p 8080:3000 --name web-container myusername/my-web-app:v1.0

# Push image to Docker Hub
docker push myusername/my-web-app:v1.0
```

4. Uploading Projects to GitHub via GitBash

Commands to push local code to a new GitHub repository from GitBash:

```
git init
git add .
git commit -m 'Initial project release'
```

```
git branch -M main  
git remote add origin https://github.com/your-user/your-repo.git  
git push -u origin main
```

KEY TAKEAWAYS & SUMMARY

- ✓ Containers share the host OS kernel, providing lightweight portability and rapid startup.
- ✓ Dockerfile defines container blueprints using directives like FROM, WORKDIR, COPY, RUN, and CMD.
- ✓ Docker Hub acts as a centralized repository registry for publishing container images.